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Linear Algebra on $\mathbb{C}^n$ (Review)

Notation

- **$m \times n$ matrix:** $A = [a_{ij}]$, where $a_{ij} \in \mathbb{C}$.
- **n -vector:** $X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$, where $x_i \in \mathbb{C}$.
- **Hermitian Transpose:** $A^\dagger$ (or A^H) denotes the complex conjugate transpose of A . If $A = [a_{ij}]$, then $A^\dagger = [a_{ji}^*]$, where a_{ji}^* is the complex conjugate of a_{ji} .
- **Norm of X :** $\|X\| = \sqrt{X^\dagger X} \geq 0$. It is zero if and only if X is the zero vector.

Positive Definite Matrices

An $n \times n$ matrix S is **positive definite** if:

$$X^\dagger S X > 0$$

for every non-zero vector X .

Note: * S is **non-negative definite** (or positive semi-definite) if $X^\dagger S X \geq 0$ for every vector X . * If A is an $m \times n$ matrix and $S = A^\dagger A$, then S is non-negative definite because:

$$X^\dagger S X = X^\dagger A^\dagger A X = (AX)^\dagger (AX) = \|AX\|^2 \geq 0$$

Eigenvalues and Eigenvectors

If S is an $n \times n$ matrix and E is an n -vector such that:

$$SE = \lambda E$$

for some scalar λ , then E is an **eigenvector** of S and λ is its associated **eigenvalue**.

Theorem: If S is positive definite (or non-negative definite), every eigenvalue of S is positive (or non-negative).

Proof: Assume $SE = \lambda E$. Left-multiplying by $E^\dagger$ gives:

$$E^\dagger SE = E^\dagger (\lambda E) = \lambda (E^\dagger E) = \lambda \|E\|^2$$

Solving for λ , we get:

$$\lambda = \frac{E^\dagger SE}{\|E\|^2} > 0$$

(The inequality holds because S is positive definite. A similar argument holds for non-negative definite matrices where $\lambda \geq 0$.)

Remarks about Eigenvalues and Eigenvectors

If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of an $n \times n$ matrix S :

(a) Determinant and Trace:

$$\det(S) = \lambda_1 \lambda_2 \cdots \lambda_n$$

$$\text{tr}(S) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$$

(b) Orthogonality: If E_1 and E_2 are eigenvectors of a Hermitian matrix S corresponding to distinct eigenvalues, then $E_1^\dagger E_2 = 0$ (i.e., E_1 and E_2 are orthogonal).

(c) Invertibility: A positive definite matrix has a positive determinant (since all $\lambda_i > 0$) and is thus invertible.

(d) Spectral Decomposition: If S is a Hermitian matrix with eigenvalues $\lambda_1, \dots, \lambda_n$ and corresponding **orthonormal** eigenvectors $E_1, \dots, E_n$, then S can be factored as:

$$S = \lambda_1 E_1 E_1^\dagger + \lambda_2 E_2 E_2^\dagger + \cdots + \lambda_n E_n E_n^\dagger$$

(e) Rank: If S has l non-zero eigenvalues, then l is the **rank** of S .

Example

Let $S = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$.

1. Check Positive Definiteness:

$$\begin{aligned} X^\dagger S X &= \begin{bmatrix} x_1^* & x_2^* \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1^* & x_2^* \end{bmatrix} \begin{bmatrix} 3x_1 + x_2 \\ x_1 + 3x_2 \end{bmatrix} \\ &= 3|x_1|^2 + x_1^* x_2 + x_2^* x_1 + 3|x_2|^2 \\ &= 3|x_1|^2 + 2\text{Re}(x_1^* x_2) + 3|x_2|^2 > 0 \quad (\text{for } X \neq 0) \end{aligned}$$

2. Find Eigenvalues: The eigenvalues are the roots of the characteristic polynomial:

$$\begin{aligned} 0 = \det(\lambda I - S) &= \det \begin{bmatrix} \lambda - 3 & -1 \\ -1 & \lambda - 3 \end{bmatrix} \\ &= (\lambda - 3)^2 - 1 \\ &= \lambda^2 - 6\lambda + 8 = (\lambda - 4)(\lambda - 2) \end{aligned}$$

So, $\lambda_1 = 4$ and $\lambda_2 = 2$. (Verification: $\text{tr}(S) = 4 + 2 = 6$, and $\det(S) = 4 \times 2 = 8$.)

3. Find Eigenvectors: For $\lambda_1 = 4$, $SE_1 = 4E_1$. Let $E_1 = \begin{bmatrix} a \\ b \end{bmatrix}$.

$$3a + b = 4a \implies a = b$$

Normalizing so that $\|E_1\| = 1$ gives $E_1 = \frac{\sqrt{2}}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

For $\lambda_2 = 2$, by orthogonality ($E_1^\dagger E_2 = 0$), it is easily determined that:

$$E_2 = \frac{\sqrt{2}}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

4. Spectral Decomposition:

$$\begin{aligned} \lambda_1 E_1 E_1^\dagger + \lambda_2 E_2 E_2^\dagger &= 4 \left(\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right) + 2 \left(\frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \right) \\ &= \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} = S \end{aligned}$$

Unitary Matrices

An $n \times n$ matrix U is **unitary** if:

$$U^\dagger U = I$$

where I is the $n \times n$ identity matrix. Note that this implies $U^\dagger = U^{-1}$, so $UU^\dagger = I$ as well.

Orthonormal Columns: Writing U in terms of its column vectors $u_1, \dots, u_n$:

$$U = \begin{bmatrix} | & & | \\ u_1 & \dots & u_n \\ | & & | \end{bmatrix}$$

Evaluating $U^\dagger U$ gives:

$$U^\dagger U = \begin{bmatrix} - & u_1^\dagger & - \\ & \vdots & \\ - & u_n^\dagger & - \end{bmatrix} \begin{bmatrix} | & & | \\ u_1 & \dots & u_n \\ | & & | \end{bmatrix} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

This demonstrates that $u_i^\dagger u_j = 1$ if $i = j$, and $u_i^\dagger u_j = 0$ if $i \neq j$. That is, the columns (and rows) of U are **orthonormal vectors**.

Eigenvalues of Unitary Matrices: If λ is an eigenvalue of U with corresponding eigenvector E :

$$UE = \lambda E$$

Left-multiplying by $U^\dagger$:

$$U^\dagger U E = U^\dagger (\lambda E)$$

Since $U^\dagger U = I$:

$$E = \lambda U^\dagger E \implies U^\dagger E = \frac{1}{\lambda} E$$

This establishes that E is also an eigenvector of $U^\dagger$, with eigenvalue $\frac{1}{\lambda}$.